

Measure DC Current in a Circuit

Current determined from a DC voltage-divider measurement using Ohm's law

Method

The divider is a series circuit, so the current through the lower resistor equals the circuit current. With a +5 V W1 input, the Scopy Voltmeter measured the voltage across the 1.1 k Ω lower resistor.

Calculation

Measured V_{R2} : 3.251 V $R2$: 1.1 k Ω

$$I = V_{R2} / R2 = 3.251 \text{ V} / 1100 \Omega = 0.00296 \text{ A} = 2.96 \text{ mA.}$$

Brief procedure

1. Set W1 to +5 V DC. 2. Measure V_{OUT} across $R2$ using 1+ at V_{OUT} and 1- at GND. 3. Apply Ohm's law to obtain the series current.

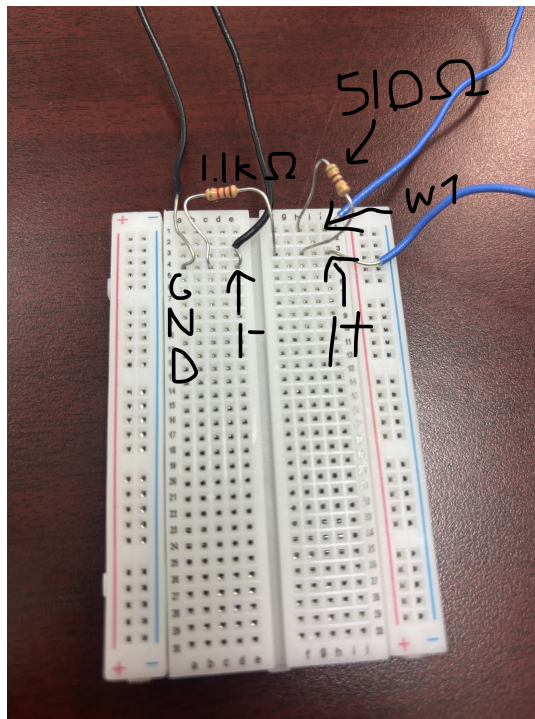


Figure 1. Labeled breadboard photo of the two-resistor series divider used for the current calculation.

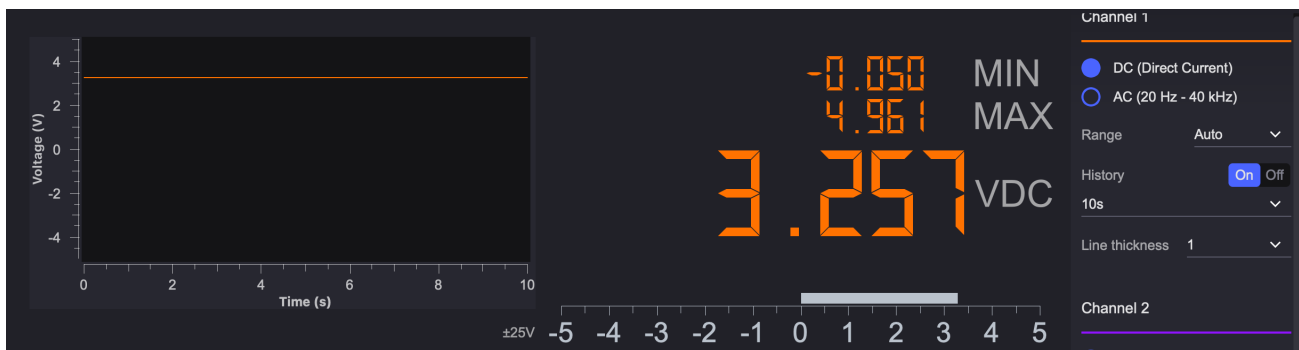


Figure 2. Scopy Voltmeter evidence: +3.251 VDC across the 1.1 k Ω lower resistor.